# MILITARY SPECIFICATION

## ATTACK CONSOLE MARK 39 MOD 0

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

## 1. SCOPE

1.1 Scope. This specification describes the requirements for the Attack Console Mark 39 Mod 0, BUORD LD 498311, a major component for the Submarine Fire Control System Mark 112 Mod 1.

## 2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation for bids, or as noted, form a part of this specification:

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-601 | Boxes, Wood, Cleated-Plywood |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-corner |

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation, Methods of |
| MIL-A-5092 | Adhesive, Rubber (Synthetic and Reclaimed Rubber Base) |
| MIL-E-5272 | Environmental Testing, Aeronautical and Associated Equipment, General Specification for |

FSC 1210

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|---|---|
| MIL-I-8835 | Indicator, Humidity, Card, Chemically Impregnated |
| MIL-G-12803 | Gasket Materials, Nonmetallic |
| MIL-I-16910 | Interference Measurement, Radio, Method and Limits; 14 Kilocycles to 1000 Megacycles |
| MIL-T-17113 | Tests, Shock, Vibration, and Inclination (for Electronic Equipment) General Specification |
| MIL-F-18870 | Fire Control, Optical and Associated Equipment Intended for Shipboard Use, The General Specification for Design and Procurement of |
| MIL-W-21927 | Weapons; Handling and Preparation for Delivery of; General Requirements for |
| MIL-I-26860 | Indicator, Humidity, Plug, Color Change |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-108 | Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-167 | Mechanical Vibrations of Shipboard Equipment |

DRAWINGS

NAVY DEPARTMENT

| | |
|---|---|
| BUORD 1876249 | Resilient Mount Assembly |
| BUORD LD 498172 | Mechanical Display Cabinet |
| BUORD LD 498175 | Electrical Control Cabinet |



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| | |
|---|---|
| BUORD LD 498288 | Firing Order Cabinet |
| BUORD LD 498305 | Ballistic Switch Cabinet |
| BUORD LD 498311 | Attack Console Mark 39 Mod 0 |
| BUORD LD 498315 | Base Cabinet |
| BUORD LD 498403 | Resilient Mount Assembly |

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

| | |
|---|---|
| NAVWEPS OD 12409 | Factory Acceptance Tests Attack Console Mark 39 Mod 0 |
| NAVORD OP 400 | General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment |
| NAVORD OS 6341 | Miscellaneous General Ordnance Design Requirements |

(Copies of specifications, standards, and drawings requested by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATION

| | |
|---|---|
| Handbook H28 and all Supplements thereto | Screw-thread Standards for Federal Services |

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)



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AMERICAN TRUCKING ASSOCIATION, INC., PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th Street, N.W., Washington 6, D.C.)

3. REQUIREMENTS

3.1 Description. The Attack Console Mark 39 Mod 0, BUORD LD 498311, is a major component of the Fire Control System Mark 112 Mod 1. It embodies several electro-mechanical analog computers and houses the equipment necessary for target analysis, position keeping, weapon order generation, and programming of firing of wire guidance, homing, and conventional torpedoes.

3.2 General requirements.

3.2.1 Preproduction sample. When specified in the contract or order, preproduction samples of the Attack Console Mark 39 Mod 0 shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification and the design. Provision of samples shall be in accordance with 4.1.1.

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials entering into the final assembly of the Attack Console Mark 39 Mod 0 shall conform to the detailed requirements indicated on the drawings and specifications listed in BUORD LD 498311.

3.2.3 Finishes, coatings, construction and dimensions. Finishes, coatings, construction and dimensions for the Attack Console Mark 39 Mod 0 shall be in accordance with the applicable drawings, specifications, and publications listed in BUORD LD 498311.

3.2.4 Moisture, fungus, and corrosion. The Attack Console Mark 39 Mod 0 shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified on the drawings and specifications listed in BUORD LD 498311.

3.2.5 Marking. Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawings listed in BUORD LD 498311.



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3.2.6 Interchangeability. Individual parts of assemblies fabricated in agreement with their applicable drawings and specifications shall be compatible with mating parts of the unit and interchangeable with other like units without selection or fitting. If greater or smaller tolerances than those shown on the drawings or specifications are required for interchangeability; or if hand fitting or selection is required, then changes in dimensions are necessary and revision and correction of the applicable drawings is required. See 3.2.10.

3.2.7 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on applicable drawings. If so stipulated in the contract, order or requisition, the Government will furnish drawings or pertinent Navy Final Inspection Gages for information and guidance in design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Naval Final Inspection Gages. (For test equipment, see 4.3.)

3.2.8 Threads. Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28 and all supplements thereto. The class or fit for threads shall be as specified on the applicable drawings and the NOT-GO gaging practice shall be as set forth in the Supplement to the aforementioned Handbook H28.

3.2.9 Requests for deviation. Requests for deviation from this specification, applicable drawings, specifications, publications, materials, and processes specified herein, shall be as set forth in OP 400.

3.2.10 Conflicting requirements. Conflicting requirements arising between this specification and any specification, publication, or drawing listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation, clarification, and resolution (or correction).

3.3 Definitions. Definitions applicable to this specification and associated drawings are as follows:

3.3.1 Tropical environment. Tropical environment is defined as conditions of high humidity, high temperature, and spore exposure, which presents an opportunity for fungi growth.

3.3.2 Inclination. Inclination is defined as the acute angle between a given plane and the true vertical plane.



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3.4 Performance requirements and product characteristics. The Attack Console Mark 39 Mod 0 shall meet the following performance requirements and product characteristics.

3.4.1 Dielectric checks. The Attack Console Mark 39 Mod 0 wiring shall show no evidence of damage when high-potential and insulation resistance tests specified in NAVWEPS OD 12409 are conducted.

3.4.2 Visual checks. All selector switches, potentiometers, toggle switches, handcranks, and knobs shall function properly when the tests specified in NAVWEPS OD 12409 are conducted. All fuses, ballistic plugs, switch tabs, cams and alignment pins shall be installed properly as specified in NAVWEPS OD 12409.

3.4.3 Power supply. The power supply output voltages shall be within the tolerances specified in NAVWEPS OD 12409 when the input voltages are at the nominal value.

3.4.4 Warning and indicator lights. All warning and indicator lights shall function properly when the electrical tests specified in NAVWEPS OD 12409 are conducted.

3.4.5 Anti-circling run limits, burst run control and steer permit indicator. The Attack Console Mark 39 Mod 0 shall function properly when the anti-circling run limits, burst run control and steer permit indication tests specified in NAVWEPS OD 12409 are conducted.

3.4.6 Own ship's motion north and east receivers. The own ship's motion north and east receiver units shall function properly when the tests specified in NAVWEPS OD 12409 are conducted.

3.4.7 Position keeper control panel. The position keeper control panel shall function properly when the tests specified in NAVWEPS OD 12409 are conducted.

3.4.8 Angle solver. The angle solver shall pass the functional tests specified in NAVWEPS OD 12409.

3.4.9 Firing order sequence. All selector switches, pushbuttons and indications in the firing order sequence portion shall function properly when the tests specified in NAVWEPS OD 12409 are conducted

3.4.10 Limit light check. The limit lights shall function properly when the tests specified in NAVWEPS OD 12409 are conducted.

3.4.11 Error meter and error light check. The meters and error lights shall function properly when the tests specified in NAVWEPS OD



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12409 are conducted.

3.4.12 Target analyzer. The lights, meters and switches on the target analyzer control panel shall function properly when the tests specified in NAVWEPS OD 12409 are conducted.

3.4.13 Action cut-out switch panel. The switches on the action cut-out switch panel shall function properly when the tests specified in NAVWEPS OD 12409 are conducted.

3.4.14 Mark 39 stop control and the wire guide continuity switch. The Mark 39 stop control and the wire guide switch shall function properly when the tests specified in NAVWEPS OD 12409 are conducted.

3.4.15 Servo response. The servo's shall function properly when the electrical tests specified in NAVWEPS OD 12409 are conducted.

3.4.16 Accuracy. The Attack Console Mark 39 Mod 0 shall solve the problems specified in NAVWEPS OD 12409 with an accuracy such that the errors shall be within the limits specified therein.

3.4.16.1 Angle solver accuracy. The angle solver accuracy shall be tested in two modes of operation.

3.4.16.1.1 Preset section accuracy. The fire control problem for the torpedo operating entirely as a preset torpedo shall be solved with average errors no greater than those specified in NAVWEPS OD 12409.

3.4.16.1.2 Dead reckoner accuracy. The dead reckoner shall be capable of solving for torpedo position with respect to the launching submarine with average errors no greater than those specified in NAVWEPS OD 12409.

3.4.16.2 Target analyzer. The accuracy of the target analyzer shall be within the limits specified in NAVWEPS OD 12409.

3.4.16.3 Position keeper. The accuracy of the position keeper shall be within the limits specified in NAVWEPS OD 12409.

3.4.16.4 Continuous position keeper. The accuracy of the continuous position keeper shall be within the limits specified in NAVWEPS OD 12409.

3.4.16.5 Sonar converter. The accuracy of the sonar converter shall be within the limits specified in NAVWEPS OD 12409.

3.4.17 Panel illumination. The panels shall be illuminated with



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white and red light as specified in NAVWEPS OD 12409.

3.4.18 Miscellaneous electrical checks. The Attack Console Mark 39 Mod 0 shall pass the tests specified in "Miscellaneous Electrical Checks" NAVWEPS OD 12409.

3.5 *Environmental requirements.* The Attack Console Mark 39 Mod 0 shall meet the other applicable requirements of this specification during and after tests defined or specified in 4.5.

3.5.1 Vibration, operating condition. The Attack Console Mark 39 Mod 0 shall be capable of withstanding the Type I vibration requirements as specified in MIL-STD-167 and shall meet all performance and product characteristics requirements of this specification during and after vibration tests and thereafter shall show no damage.

3.5.1.1 Vibration-transportation. The Attack Console Mark 39 Mod 0 after being subjected to tests specified in MIL-W-21927 shall show no damage and shall meet all the performance and product characteristics requirements of this specification.

3.5.2 Shock, operating condition. The Attack Console Mark 39 Mod 0 shall be capable of withstanding the Class A shock requirements as specified in MIL-T-17113 and shall meet all performance and product characteristics requirements of this specification before and after shock tests and thereafter shall show no damage.

3.5.2.1 Shock-transportation, storage and handling. The Attack Console Mark 39 Mod 0 after being subjected to tests specified in MIL-W-21927 shall show no damage and shall meet all the performance and product characteristics requirements of this specification.

3.5.3 Temperature and humidity. The Attack Console Mark 39 Mod 0 shall be capable of withstanding operational ambient temperature and humidity conditions specified in MIL-F-18870 and storage temperatures in accordance with Condition II of NAVORD OS 6341, and thereafter shall show no corrosion or damage and shall meet the performance and product characteristics of this specification.

3.5.4 Tropical environment. The Attack Console Mark 39 Mod 0 shall be capable of withstanding the tropical environmental conditions specified in MIL-E-5272 under "Fungus Resistance Tests" and thereafter shall show no fungus growth, corrosion or other damage, and shall meet the performance and product characteristics of this specification.

3.5.5 Inclination. The Attack Console Mark 39 Mod 0 shall be capable of withstanding the inclination requirements as specified in MIL-



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T-17113 and shall meet all performance and product characteristics requirements of this specification during and after inclination tests and thereafter shall show no damage.

3.5.6 Enclosure. The Attack Console Mark 39 Mod 0 shall be of spraytight construction as defined in MIL-STD-108.

3.5.7 Illumination. Dial illumination shall conform with performance requirements of "illumination for dark adaption" as specified in MIL-F-18870.

3.5.8 Radio frequency and low frequency noise. The maximum radiated and the maximum conducted noise-interference and electrical disturbance shall not exceed the values specified in MIL-I-16910.

3.5.9 Power system variations. The equipment shall function properly under the transient and power variability conditions specified in MIL-F-18870 under the heading "Power Availability".

3.5.10 Mechanical noise and vibration. The noise generated by the Attack Console Mark 39 Mod 0 shall be such that the arithmetic average of the sound pressure levels in the four octave bands of 300-600, 600-1200, 1200-2400, and 2400-4800 cycles per second shall not exceed 65 decibels.

3.6 Clean-up. Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.7 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing and freedom of parts from burrs and sharp edges.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are



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deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Preproduction sample. When specified in the contract or order, a lot of 2 Attack ConsolesMark 39 Mod 0 shall be manufactured and submitted to the Government inspector who shall examine the sample to verify strict conformance with drawings and specifications prior to production tests at an activity designated in the contract or by the procuring activity. Further production of the item by the contractor prior to approval of the contracting activity or the completion of the tests on the preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.2 Periodic production samples. Samples for periodic production testing may be randomly selected by the Government inspector in accordance with MIL-STD-105, Inspection Level L-4, for submission to a Bureau of Naval Weapons assigned activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The Attack Console Mark 39 Mod 0 shall be considered satisfactory if the Attack Console meets all specifications and pertinent drawing requirements and withstands the tests as specified in 4.4 and 4.5 inclusive.

4.2 Classification of test. The inspection and testing of the Attack Console Mark 39 Mod 0 shall be classified as follows:

4.2.1 Acceptance tests. These tests are to be accomplished on each Attack Console Mark 39 Mod 0 being submitted for acceptance under contract. Acceptance tests shall be either performed by the manufacturer and witnessed by the Government inspector, or these tests may be conducted by the Government inspector. These tests are detailed in 4.4.

4.2.2 Environmental tests. Environmental tests are to be performed on equipments selected by the procuring activity and will be performed at the expense of the Government at a time and place to be designated by the procuring activity. The contractor shall provide a representative for technical advice and assistance, if requested by the procuring activity. Environmental tests are detailed in 4.5. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for the equipment shall be corrected by the contractor on all units procured on the contract at no expense to the Government, provided that the contractor is notified of such defects within the guarantee period of the contract. The need for environmental tests shall be as required by the procuring agency.



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4.2.3 Test after delivery. Unless otherwise specified in the contract, the Government reserves the right to conduct any or all the tests specified herein after delivery. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for the equipment shall be corrected by the contractor on all units procured on the contract at no expense to the Government, provided the contractor is notified of such defects within the guarantee period of the contract.

4.3 Test equipment. The following dependent equipments and items of test equipment are required to perform the acceptance tests set forth in this specification:

4.3.1 Dependent equipments.

4.3.1.1 Amplifier Mark 140 Mods 0, 1 or 2

4.3.1.2 Switch Box Mark 5 Mod 2

4.3.1.3 Indicator Panel Mark 21 Mod 0 or 2

4.3.2 Special test equipment.

4.3.2.1 Shop accessory SA 2530404, Dummy Director Test Fixture.

4.3.2.2 Shop accessory SA 2530411, Bearing Memory Test Fixture.

4.3.2.3 Shop accessory SA 2530414, Dead Reckoning Simulator Test Fixture.

4.3.3 Standard test equipment.

4.3.3.1 Power sources

4.3.3.1.1 115-volts alternating current plus or minus 10 percent, 400 cycles-per-second plus or minus 6 percent, 3-phase, 1250-volt-amperes, 90 percent power factor, 6 percent maximum harmonic.

4.3.3.1.2 26 volts direct current plus or minus 10 percent, 10 amperes, 15 percent maximum ripple.

4.3.3.2 Measuring equipment.

4.3.3.2.1 A high-potential insulation tester rated at 115 volts alternating current, 60 cycles-per-second, and 600 volt-amperes; capable of delivering potentials of 0 volts to 2500 volts. Means shall be provided for varying the test voltage from maximum to minimum value.



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An indicator shall be provided for indicating the presence of a one milliampere drain under test.

4.3.3.2.2 An insulation-resistance meter capable of measuring at least 3 megohms at a potential of 500 volts direct current.

4.3.3.2.3 A general purpose low frequency oscilloscope with a frequency response of from direct current to down not more than 40 percent at 100 kilocycles-per-second, an input impedance of approximately 500,000 ohms and a "y" axis sensitivity of 25 peak-to-peak millivolts-per-inch.

4.3.3.2.4 A 400 cycles-per-second alternating current voltmeter capable of measuring from 0 volts to 150 volts. The input impedance shall be approximately 5000 ohms and the accuracy shall be plus or minus 0.75 percent of full scale deflection.

4.3.3.2.5 A direct current voltmeter capable of measuring from 0 volts to 3 volts, 0 volts to 150 volts, and 0 volts to 300 volts. The input impedance shall be 5000 ohms per volt and the accuracy shall be plus or minus 0.5 percent of full scale deflection.

4.3.3.2.6 A direct current voltmeter capable of measuring from 0 volts to 7.5 volts, 0 volts to 30 volts, and 0 volts to 75 volts. The input impedance shall be 5000 ohms per volt and the accuracy shall be plus or minus 0.5 percent of full scale deflection.

4.3.3.2.7 A vacuum-tube-voltmeter for checking null voltage capable of measuring from 0.005 volts to 300 volts alternating current, at an accuracy of plus or minus 3 percent, at frequencies from 10 cycles per second to 100 kilocycles per second, with an input impedance of 2 megohms or greater.

4.3.3.2.8 A dummy synchro transmitter to transmit 0- to 90-volts alternating current, 400 cycles-per-second input signals at 1 and 10 speeds with an accuracy of plus or minus 4 minutes and shall be capable of voltage level and phase adjustment.

4.3.3.2.9 A phase meter capable of reading 0 to 90 degrees at an accuracy of plus or minus one-half degree.

4.3.3.2.10 A frequency meter capable of measuring 60 cycles-per-second at an accuracy of plus or minus one-quarter cycle.

4.3.3.2.11 A frequency meter capable of measuring 400 cycles-per-second at an accuracy of plus or minus 1 cycle.



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4.3.3.2.12 A timer capable of measuring 1 hour intervals at an accuracy of plus or minus one-half second.

4.3.3.2.13 A low voltage frequency meter capable of measuring 400 cycles-per-second at an accuracy of plus or minus 1 cycle.

4.4 Acceptance tests

4.4.1 The contractor or Government inspector shall conduct acceptance tests as specified in 4.2.1 to assure that the Attack Console Mark 39 Mod 0 is in compliance with the requirements of 3.2, 3.4, 3.6 and 3.7.

4.4.2 Test conditions. Unless otherwide specified, the Attack Console Mark 39 Mod 0 shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature. Room ambient 65 degrees Fahrenheit to 95 degrees Fahrenheit (18 degrees centigrade to 35 degrees centigrade).

4.4.2.2 Altitude. Normal ground atmospheric pressure.

4.4.2.3 Vibration. None.

4.4.2.4 Humidity. Room ambient to 95 percent relative maximum.

4.4.3 Test and inspection equipment and facilities. The Government will furnish the dependent equipment specified in 4.3.1. The manufacturer shall furnish and maintain all necessary standard and special test equipment and facilities and personnel for performing all acceptance tests. If specified in the contract, the Government may furnish the special test equipment specified in 4.3.2 when it is available. The standard and special test equipment shall be adequate in quantity and, unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.4 Test procedure.

4.4.4.1 Dielectric. Conduct high-potential and insulation resistance tests in accordance with NAVWEPS OD 12409 "Dielectric Tests" to determine compliance with the requirements specified in 3.4.1.

4.4.4.2 Visual checks. Conduct visual checks in accordance with NAVWEPS OD 12409 to determine compliance with the requirements specified in 3.4.2.



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4.4.4.3 Power supply. Test the output of the power supply in accordance with NAVWEPS OD 12409 "Electrical Checks" to determine compliance with the requirements specified in 3.4.3.

4.4.4.4 Warning and indicator lights. Conduct the warning and indicator light tests in accordance with NAVWEPS OD 12409 "Handcrank Warning Lights, Speed Unit Selector Switch Lights, Set Light Checks, Mode Indicator Lights" to determine compliance with the requirements specified in 3.4.4.

4.4.4.5 Anti-circling run limits, burst run control and steer permit indicator. Conduct anti-circling run limits, burst run control and steer permit indicator tests in accordance with NAVWEPS OD 12409 to determine compliance with the requirements specified in 3.4.5.

4.4.4.6 Own ship's motion north and east receiver. Conduct own ship's motion north and east receiver checks in accordance with NAVWEPS OD 12409 to determine compliance with the requirements specified in 3.4.6.

4.4.4.7 Position keeper control panel. Conduct position keeper control panel functional tests in accordance with NAVWEPS OD 12409 "Electrical Checks" to determine compliance with the requirements specified in 3.4.7.

4.4.4.8 Angle solver. Conduct angle solver functional tests in accordance with NAVWEPS OD 12409 to determine compliance with the requirements specified in 3.4.8.

4.4.4.9 Firing order sequence. Conduct firing order sequence tests in accordance with NAVWEPS OD 12409 to determine compliance with the requirements specified in 3.4.9.

4.4.4.10 Limit light check. Conduct limit light checks in accordance with NAVWEPS OD 12409 to determine compliance with the requirements specified in 3.4.10.

4.4.4.11 Error meter and error light check. Conduct error meter and error light checks in accordance with NAVWEPS OD 12409 to determine compliance with the requirements specified in 3.4.11.

4.4.4.12 Target analyzer. Conduct target analyzer control panel functional tests in accordance with NAVWEPS OD 12409 to determine compliance with the requirements specified in 3.4.12.

4.4.4.13 Action cut-out switch panel. Conduct action cut-out



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switch tests in accordance with NAVWEPS OD 12409 to determine compliance with the requirements specified in 3.4.13.

4.4.4.14 Mark 39 stop control and wire guide continuity switch. Conduct Mark 39 stop control and wire guide continuity switch tests in accordance with NAVWEPS OD 12409 to determine compliance with the requirements specified in 3.4.14.

4.4.4.15 Servo response. Conduct servo response tests in accordance with NAVWEPS OD 12409 "Input and Output Transmission Checks" to determine compliance with the requirements specified in 3.4.15.

4.4.4.16 Accuracy. Conduct accuracy tests in accordance with NAVWEPS OD 12409 "Accuracy Checks" to determine compliance with requirements specified in 3.4.16.

4.4.4.17 Panel illumination. Conduct the panel illumination tests in accordance with NAVWEPS OD 12409.

4.4.4.18 Miscellaneous electrical checks. Conduct the miscellaneous electrical checks "Spread Order Check, Stratum Warning Check, Torpedo Room Control Light Check and System Timer" checks in accordance with NAVWEPS OD 12409

4.5 Environmental tests.

4.5.1 Vibration, operating condition. The Attack Console Mark 39 Mod 0, rigidly mounted in a suitable vibration fixture, shall be subjected to Type I vibration tests specified in MIL-STD-167 to determine compliance with requirements specified in 3.5.1 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.1.1 Vibration-transportation. The Attack Console Mark 39 Mod 0 packaged and packed as specified in Section 5, shall be vibration tested as specified in MIL-W-21927 to determine compliance with requirements specified in 3.5.1.1.

4.5.2 Shock, operating condition. The Attack Console Mark 39 Mod 0, rigidly mounted in a suitable shock fixture, shall be subjected to Class A shock tests specified in MIL-T-17113 to determine compliance with requirements specified in 3.5.2 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.2.1 Shock-transportation, storage and handling. The Attack Console Mark 39 Mod 0 packaged and packed as specified in Section 5,



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shall be subjected to the rough handling tests specified in MIL-W-21927 to determine compliance with requirements specified in 3.5.2.1.

4.5.3 Temperature and humidity. The Attack Console Mark 39 Mod 0, shall be subjected to operational ambient temperature and humidity tests specified in MIL-F-18860 and storage temperatures in accordance with Condition II of NAVORD OS 6341 to determine compliance with requirements specified in 3.5.3 and thereafter shall meet the performance requirements and product characteristics of this specification.

4.5.4 Tropical environment. The Attack Console Mark 39 Mod 0, shall be exposed to Fungus Resistance Tests specified in MIL-E-5272 to determine compliance with requirements specified in 3.5.4 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.5 Inclination. The Attack Console Mark 39 Mod 0, shall be subjected to inclination tests specified in MIL-T-17113 to determine compliance with requirements specified in 3.5.5, and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.6 Enclosure. The degree of spraytight enclosure shall be tested in accordance with MIL-STD-108 to determine compliance with requirements specified in 3.5.6.

4.5.7 Illumination. The Attack Console Mark 39 Mod 0, shall be tested for "illumination for dark adaption" as specified in MIL-F-18870 to determine compliance with requirements specified in 3.5.7.

4.5.8 Radio frequency and low frequency noise. The Attack Console Mark 39 Mod 0, shall be tested for radio frequency and low frequency noise interference in accordance with MIL-I-16910 to determine compliance with requirements specified in 3.5.8.

4.5.9 Power system variations. The Attack Console Mark 39 Mod 0, shall be subjected to transient and power system variability conditions specified in MIL-F-18870 under the heading "Power Availability" to determine compliance with requirements of 3.5.9 and thereafter shall meet all the performance requirements and product characteristics of this specification.

4.5.10 Mechanical noise and vibration. The noise level tests shall be conducted in a soundproof room in accordance with the instructions given in MIL-F-18870 to determine compliance with requirements specified in 3.5.10.



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4.6 Failure of the Attack Console Mark 39 Mod 0 to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. The exterior surfaces only of the Attack Console shall be cleaned in accordance with process C-1 of Specification MIL-P-116.

5.1.1.2 Drying. Unless otherwise specified in the contract or order, the selection of the drying procedures to be used shall be at the option of the contractor, provided the procedure selected will not cause damage to the Attack Console.

5.1.1.3 Preservative application. None required.

5.1.1.4 Unit packaging. The Attack Console shall be packaged in eight (8) separate units. The contents of each package shall be:

(1) Mechanical Display Cabinet BUORD LD 498172
(2) Electrical Control Cabinet BUORD LD 498175
(3) Firing Order Cabinet BUORD LD 498288
(4) Ballistic Switch Cabinet BUORD LD 498305
(5) Base Cabinet BUORD LD 498315
(6) Interconnection Cables and Miscellaneous Hardware
(7) Accessory Box (Ballistic Plugs, Cams, Tabs)
(8) Shock Mounts BUORD LD 498493 and BUORD Drawing 1876249

Units (1), (2) and (5) shall be individually packaged in accordance with Method IIa of Specification MIL-P-116. Movable parts, susceptible to damage, shall be cushioned, secured, blocked and braced to prevent movement prior to packaging. Material for gaskets used to fabricate the package shall conform to Specification MIL-G-12803, Type 1, Class 2, Group 1, Grade 1. The adhesive for cementing the gaskets shall conform to Specification MIL-A-5092, Type III. The humidity indicator for the package shall conform to Specification MIL-I-26860, Type II. Units (3), (4) and (7) shall be individually packaged in accordance with Method IIb of Specification MIL-P-116. The humidity indicator shall conform to MIL-I-8835 (MS20003-2). The desiccant and humidity indicator shall be placed inside the accessory box of Unit (7). The accessory box shall be the inner container of the container, barrier, container of Method IIb package.



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Unit (6) shall be individually packaged in accordance with Method IA-15 of Specification MIL-P-116.

Unit (8) shall be packaged in accordance with Method III of Specification MIL-P-116.

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5%. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. Not applicable.

5.1.3 Level C.

5.1.3.1 Cleaning, drying and preservative application. The cleaning, drying and preservative application of the Attack Console shall be in accordance with 5.1.

5.1.3.2 Unit packaging. The Attack Console shall be individually packaged in accordance with the manufacturer's commercial practice. The packaging shall afford adequate protection against corrosion, deterioration and physical damage during direct shipment from supply source to the first receiving activity for immediate use. The method used shall be commensurate with shipping at the lowest practical commercial rates.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Units (1) through (5) packaged as described in 5.1, shall be individually packed in (overseas type) wood cleated plywood or nailed wood box, conforming to Specification PPP-B-601, Style A or B, Type 3 load or PPP-B-621, Class 2, Style 2, Type 3 load, respectively. The unit components of the Attack Console shall be cushioned, blocked and braced to provide protection from damage during transportation, handling and storage. The boxes for Units (1), (2) and (5) shall be modified to include a skid base with rubbing strips and interior diagonal bracing. The boxes for Units (1), (2) and (5) shall be provided with an inspection port for viewing the humidity indicator.

Units (6), (7) and (8) packaged as described in 5.1, shall be packed in a nailed wood box, conforming to Specification PPP-B-621, Class 2, Style 2. The pack shall incorporate sufficient cushioning material, blocking and/or bracing to provide protection from damage to the items during transportation, handling and storage.

5.2.2 Level B. Not applicable.



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5.2.3 Level C. The Attack Console, packaged as described in 5.1, shall be individually packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest rate. Containers shall be in accordance with the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Normal marking. In addition to any special marking required by the contract or order, the unit packages and shipping containers shall be marked in accordance with requirements of Military Standard MIL-STD-129, including the serial number of the Attack Console contained therein.

5.3.2 Special marking. The following special marking shall appear on the top, one (1) side and one (1) end of each shipping container:

DELICATE EQUIPMENT
HANDLE WITH CARE

6. NOTES

6.1 Intended use. The Attack Console Mark 39 Mod 0 is intended to be used in the Submarine Fire Control System Mark 112 Mod 1.

6.2 Ordering data. Procurement documents should specify the following:

(a) The specific title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) The exceptions to this specification, applicable drawings, and other documents.
(d) The guarantee period (see 4.2.2).

6.3 Criteria for use of levels of preservation, packaging and packing.

6.3.1 Preservation and packaging levels.

6.3.1.1 Level A. The degree of preservation and packaging used for this level shall afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.3.1.2 Level B. The degree of preservation and packaging used for this level shall afford adequate protection against known conditions



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which are less hazardous than Level A is designed to meet. This requires a higher degree of protection than afforded by Level C preservation and packaging. The use of Level B should be based on firmly established knowledge of the shipment and storage conditions to be encountered and a determination that monetary savings will result.

6.3.1.3 Level C. The degree of preservation and packaging used for this level shall afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

6.3.2 Packing levels.

6.3.2.1 Level A. The degree of packing used for this level shall afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.3.2.2 Level B. The degree of packing used for this level shall afford adequate protection against damage during multiple shipments. These packs shall be designed to be shipped and handled under cover and stored in warehouses or other structures affording equivalent protection from weather.

6.3.2.3 Level C. The degree of packing used for this level shall afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level, in general, will conform to the applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

NOTICE

When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specification or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.



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